

Lecture 02-part 2: Literature Search + Review

Marwa Mahmoud

Slides prepared by : Joseph O'Hagan and Marwa Mahmoud

Today's lecture

- Highlight the key parts of the readings
- “Primary vs Secondary Research”
- Taxonomy of scientific articles
- Conference vs Journal publications in Computer Science
- The different types of “research paper/article”
- How to find “good quality” research papers
- Discuss the written assessed exercise

The different types of research article / publication

As a researcher you can publish in a variety of ways

- **Journal papers**
- **Full conference paper**
- **Short conference paper**
- **Late-breaking work / poster paper**
- **Demonstration paper**
- **Workshop paper**

When you find a paper you need to know what you are looking at

Journal papers

Computer science and may be interdisciplinary work

- Bulkier than conference papers (e.g. 20+ pages)

A number of studies presented together to present a research vision

- More time intensive to do this work and get it published
- Multidisciplinary work e.g. AI in digital health

Conference papers

Either full (6-12 pages) or short (4-6) paper

- The predominant form of publication in computer science
- As a researcher most of your time will be spent producing these
- The scope of the paper varies considerably

Shorter submission-publication time than journal papers
(7 months vs 24 months)

Late-breaking works / Poster paper

Non-archival works (2-6 pages)

- Designed as a low-cost way to present work and get feedback
- Presented in a poster presentation session at a conference

Easier review process than a full paper (so easier to get accepted)

Seen as less prestigious than a full paper

Demonstration papers

Summarising a demonstration you run at a conference (2-6 pages)

- Allows attendants to try system people have built
- Another way of getting feedback



Photo from demonstration session at CHI 2019

Workshop papers

- Same format as conference papers but varies depending on the conference
- Based on the theme/topic of the workshop
- The quality of the paper depends on the quality of the workshop and if hosted in a big conference (check ranking as well) , however in general workshop papers are not as strong as main conference papers but could be more specific to certain topic

First step in research is:

Literature review!

So how to find good papers?

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Searching for papers

There are a number of ways you can find out about papers:

- Word of mouth / recommendations
- Knowing the field
- Searching for keywords and concepts
- Following authors
- Trail search
- Citation search

Searching for papers - Recommendations

Researchers remember papers they like / find interesting

- Supervisors will hand new PhD students a list of papers to read
- In a research community, you want to be in a culture where people are recommending each other works to look at

Searching for papers - Knowing the field

Identify 3 thing:

- Key publication venues in the topic (e.g. the A* conferences)
- Key researchers in the topic (who did the ground-breaking work)
- Key research groups in the topic

Last point: universities / companies have research groups focused on different specialities

Searching for papers - Knowing the field

Suppose you want to know about “research on using AR/MR/VR in transit”

- Steve Brewster leads the ViAJeRo project here at Glasgow
- A multi-year project involving many lecturers, postdocs, PhD students all working on this topic/direction of research

Approximately 25 publications so far focused on this topic alone

Searching for papers - Knowing the field

Suppose you want to know about “research on affective computing”

- Rosalind Piccard – MIT. Groups in Cambridge and CMU as well
- Hard to identify the names when you are new to the field
- Important to know how to search

Searching for papers - Keywords / Concepts

There are various archives of papers:

- ACM Digital Library
- IEEE Xplore

These are the archives for ACM and IEEE published papers respectively

Both have search engine tools built in

Searching for papers - Keywords / Concepts

There are various 'search engines' for research papers:

- Google Scholar
- DBLP

These are designed to allow you to search for papers published at any venue

- And to provide misc statistics about the paper and author

Searching for papers - Keywords / Concepts

There are various 'search engines' for research papers:

- Google Scholar
- DBLP

These search engines bring loads of papers, look for quality metrics: where the paper was published, journal/ conference scope, how many citations vs when it was published, ... etc

Searching for papers - Following authors

Researchers typically “stick to what they know”

- They specialised in specific directions of work
- E.g. a HCI VR researcher is unlikely to ever publish something about an FPGA thingy

As you read papers you will encounter authors whose work you like

Searching for papers - Following authors

Therefore it makes sense to:

- Read/check an author's back catalogue of papers
- Follow / read any new papers published by the author
- Check the back catalogue of people the author has collaborated with

A note about time

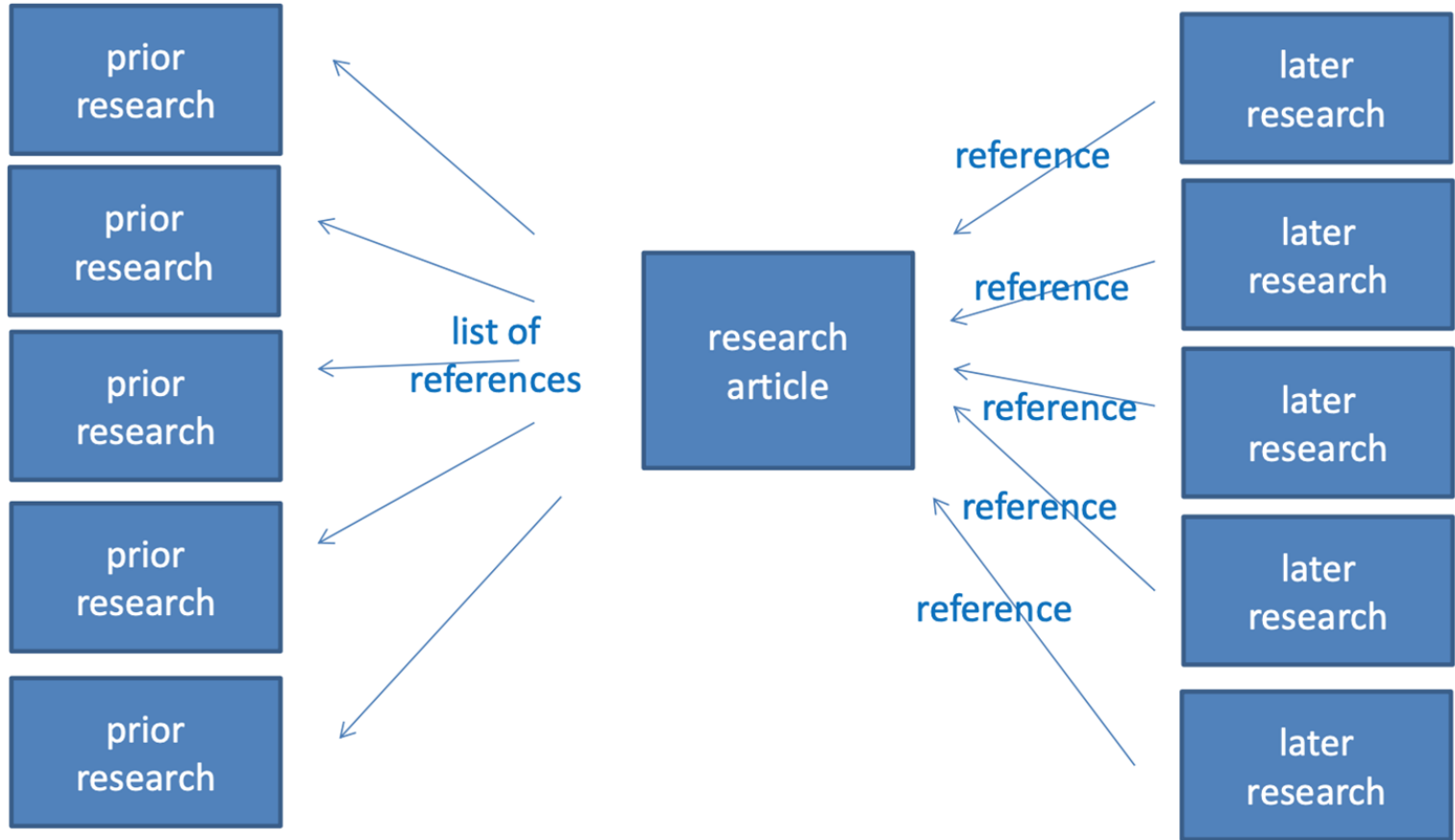
A research paper captures a moment in time

In many ways it represents all knowledge up until this point which you are now trying to expand/extend forward

- You reference/cite prior work to make your argument
- Future work reference/cites your work to make their argument

A recursive / cyclical process

“If I have seen further, it is by standing on the shoulders of giants”



We can use this “moment of time” to find papers

Can look backwards – trail search

- **To see what motivated/inspired this paper**

Can look forwards – citation search

- **To see what work has built upon this paper**
- **Maybe it's flawed and all future works trash the idea?**

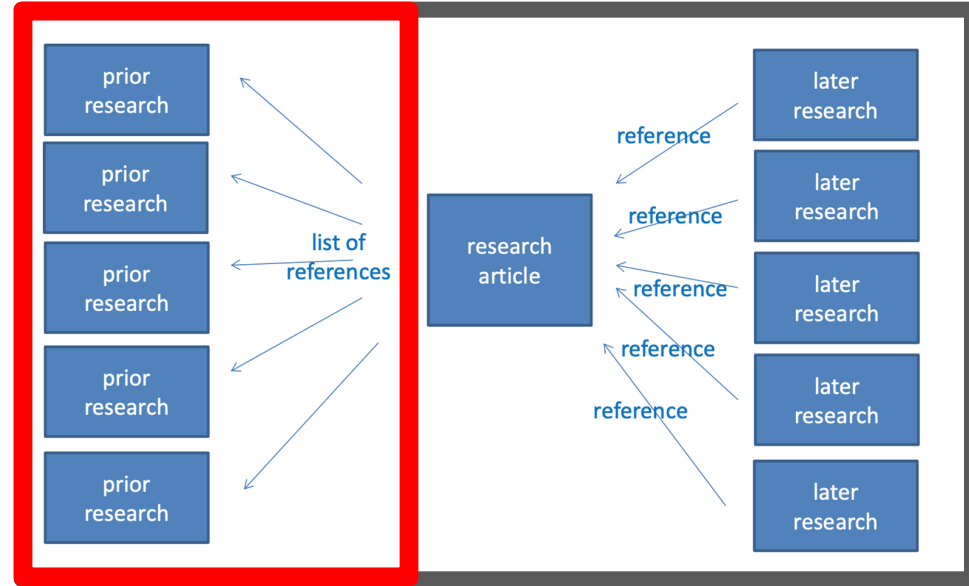
A trail search

Start with a relevant source (e.g. keyword search to find a paper)

Find a paper

Identify other relevant articles in the list of references

Repeat

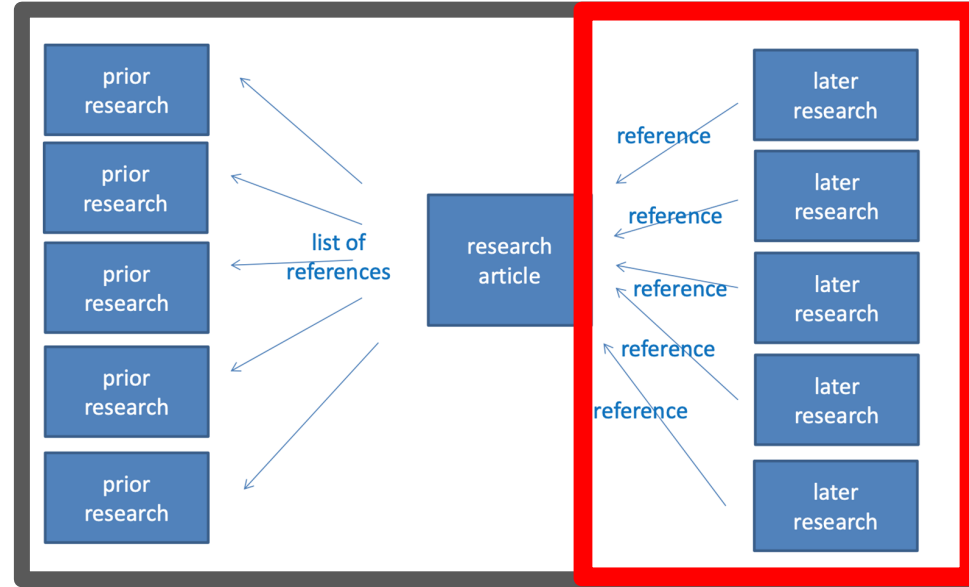


A citation search

Start with a relevant source (e.g. keyword search to find a paper)

Find a paper

Use a citation index to find sources which have cited the work



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The written assessed exercise

The written assessed exercise (20%)

Task: The task of this exercise is for you to conduct a literature search and critique of 5 research papers in Computing Science in one of these research topics:

**These are high-level
general research topics
with example sub-topics
- pick 5 papers in
any sub-topic within this**

- Facial expression recognition (e.g., state-of-the art models, challenges, application areas, ethical concerns, etc.)
- Artificial intelligence in healthcare (e.g., Interaction experimental designs, methods for classifying healthcare conditions, approaches for automatically detecting behaviour for health diagnostics, etc.)
- Computational models in chatbot applications (e.g., modelling language input and feedback, comparisons between different chatbot systems, application areas, challenges, etc.)
- Computer vision and deep learning for pedestrian detection. (e.g., methods using DNNs or similar for detecting and classifying pedestrians, application areas, smart cities, etc.).
- Face and gesture synthesis (e.g. face synthetic datasets, GANs and similar methods for synthesising facial expressions, ethical concerns, etc.

The written assessed exercise (20%)

You choose the topic / sub-topic

You choose the papers

- **Selecting appropriate topics/papers is part of the challenge**

Note: They should be peer-reviewed papers (not arxiv), at least 6 pages in length

The written assessed exercise (20%)

Your task is to produce a summary/critique of each paper using the template provided

There is a template – follow it!

- **Conference submissions have templates/word counts that you are required to follow otherwise your submission is “desk rejected”**

Failure to follow the template/instructions = a 2-band grade penalty

The written assessed exercise (20%)

You are also tasked to write a “reflective paragraph”

- **150 words on which question in the template you found the most difficult and why. And what have you learnt from this exercise.**
- **There's some education theory on why reflection is beneficial**
- **It should not be generic!**

Submission

- This is a group project, **one submission – group report** - per group. Only group leader need to submit this group report
- (Exception: if you are group of 4 people, you can review 4 papers only instead of 5)
- **In addition** to the group report submission. **Every student need to submit an individual report** – Very short - stating every team member contribution to the project (ideally it should be equal participation)

Tips to think about

Reading a 6 page paper requires less time than reading a 20+ page journal paper

- For the purpose of this assignment how is your time best spent?
- The decision is entirely yours
- Choose your papers wisely – related to the topic not general
- In computer science, the correct scope, good quality

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Summary

Readings from RMT course:

- If you haven't read them – make sure you do
- If you have – make sure you think about the things pointed out today

Summary – Primary vs secondary research

Know the difference between primary and secondary research

Know the advantages/disadvantages of each approach

Know examples of fields/topics where one might be more applicable than

Summary – Taxonomy of research papers

Know that not every paper will change the world

- Have some idea of how much science is produced every day now

Know about the different types of research paper scope:

- Breakthrough vs Ground-breaking, etc

Summary – Conferences & Papers

Know that computer science is a conference-oriented field

- And that conferences, like papers, have different levels of prestige

Know about the different types of research article and their purpose

- Full conference paper vs poster paper, etc

Summary– Finding papers

Know that a research article captures a specific moment in time

- Everything that is known at this point and an attempt to boldly go where no paper has gone before

Know different techniques for finding papers

- Citation trails, searching author publication records, etc

Employ these techniques when finding papers for your assessed exercise!

Next Week:

**The Scientific Method, Hypothesis, Ethical
Research (Ilyena Hirskyj-Douglas)**